Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	No Idea
Project / Solution Name	NexPath AI
Name (Leader)	Gan Yu Xuan
Email & Phone (Leader)	yuxuan0118.yx@gmail.com 013-404 5395
University Affiliation(s)	Universiti Teknologi Malaysia, Universiti Malaya
Live Prototype URL	https://nex-path-ai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/jasmineyh/NexPath-AI.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

NexPath AI is structured as a browser-based application. A React/Vite interface collects profile details and uploaded résumé content, presents editable extracted fields, and displays career recommendations, interactive skills visualisations and learning-roadmap cards. A validation layer ensures users confirm résumé facts before they are used for matching or résumé rewriting.

The application compares verified skills against a structured catalogue of roles and required competencies. Where configured, a backend service passes constrained prompts and permitted profile fields to an AI inference endpoint, proposed through RunPod, and returns structured suggestions for display. The frontend visualises gaps and lets users inspect explanations rather than accepting generated claims uncritically.

Replit and VS Code are used in the development workflow. The actual database, hosting configuration, AI model, persistence mechanism and deployed inference connection must be confirmed against the working repository and service dashboards. If the live demo uses mock data or local-only processing, the final declaration must explicitly state this instead of implying production integrations.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript ~5.9.3	Main language (frontend and backend)	Apache-2.0
Node.js [v]	Backend runtime	MIT
React [v]	Frontend UI	MIT
Vite [v]	Frontend build and dev server	MIT
Tailwind CSS [v]	Styling	MIT

Name & Version	Purpose in Project	License
wouter [v]	Frontend routing	Unlicense
TanStack React Query [v]	Data fetching and caching	MIT
Express 5 [v]	API server	MIT
Zod [v]	Request/response schema validation	MIT
Orval [v]	Generates React Query hooks and Zod schemas from OpenAPI	MIT
OpenAPI 3 spec	API contract (openapi.yaml)	Apache-2.0
pnpm workspaces [v]	Package and monorepo management	MIT
Prettier ^3.9.10	Code formatting	MIT
@replit/connectors-sdk ^0.4.3	Replit platform integration	[?] check on npm
Python + sentence-transformers [?]	Embedding worker on RunPod	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Embedding endpoint (own RunPod serverless endpoint, all-MiniLM-L6-v2)	Semantic skill matching	sponsor-provided credits

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts the web app and API server (two workflows)	plan
RunPod	GPU compute for the embedding endpoint	plan
GitHub	Source code hosting (Note: No database or authentication service is used.)	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2	Hugging Face, sentence-transformers/ all-MiniLM-L6-v2	Pre-trained	Apache-2.0

(Note: The deterministic keyword matcher is rule-based code, not an AI model.)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
skills.json, occupations.json, courses.json, trends.json, scenarios.json, sdg_mappings.json	Authored by the team in artifacts/api-server/data	Self-collected	Apache-2.0 (repo)	N
ESCO (structure only, not imported)	https://esco.ec.europa.eu/	Public (used as loose reference)	ESCO terms (open reuse with attribution)	N
Demo persona “Alex Tan”	Created by the team	Synthetic	Apache-2.0 (repo)	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Building the app and code	major
Claude (Anthropic)	README and submission document drafting	minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated code was reviewed by team members before use. We ran the full TypeScript typecheck (pnpm run typecheck), manually tested the main user flows in the deployed app, and checked matching results for sensible careers. Reference data was checked against its cited sources, and unverified course links are labelled as search links. The team takes full responsibility for all submitted code and content.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud credits / AI credits	Hosting, deployment and AI-assisted development
RunPod	GPU credits	Running the embedding endpoint

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No. Replit gave us fast hosting and development, but the app runs on any Node host. RunPod GPU credits power the optional semantic embedding endpoint. If it is unavailable, the app automatically falls back to the rule-based matcher, so career matching, roadmaps and scenarios still work. Both resources are replaceable.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

NexPath AI has no accounts and no database. The user's profile, chosen career, roadmap progress and resume edits are stored only in the browser's localStorage and remain on their device until they clear it. The backend is stateless and does not store profile data. When the optional embedding service is enabled, only short skill or interest phrases are sent over HTTPS to our own RunPod endpoint to compute similarity; no names, emails or contact details are sent. The endpoint key is kept as a server-side environment variable, never in the code or browser. Demo data uses a synthetic persona ("Alex Tan").

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The embedding model is small and mainly English-focused, so it may miss nuance or reflect biases from its training data. Matching covers only 11 curated occupations. Scores are suitability indicators, not predictions, and scenarios are illustrative, not forecasts. The embedding service is a single-provider dependency and may time out or cold-start. Mitigations: an 8-second timeout with automatic fallback to the rule-based matcher (retry after 30 seconds); blending scores by taking the higher value so keyword hits are never lost; showing matched and missing skills with plain-language reasons; labelling inferred strengths as system inferences; and citing sources with a last-updated date on trend entries.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Gan Yu Xuan

[24-10-2026]